

Uncertainty and sentiment in ARK portfolio firms' SEC filings

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1. Sample and research question

I examine negative language and uncertainty in 1,002 SEC filings from 60 firms. The sample includes 248 annual reports and 754 quarterly reports filed in 2021-2025. I test changes in language within firms, the association between uncertainty and subsequent volatility, and the association between negative tone and four-day filing returns.

I start with all 226 positions in the six-fund snapshot, covering 130 distinct security records. I match the original ticker, security identifier and issuer name to reviewed SEC identities. Airbus (AIR) remains separate from AAR; Discovery (DSY) remains separate from Dassault Systemes (DSY FP). Neither foreign DSY holding maps to the US-listed Big Tree Cloud. The appendix lists each excluded security.

Table 1. Holding reconciliation and filing filters

Holding disposition	Securities
Fund holdings	4
No verified SEC issuer match	3
Verified issuer, no 2021-2025 10-K/10-Q	31
Eligible listings	92

The 92 eligible listings represent 91 issuers; GOOG and GOOGL share a CIK. The filing waterfall counts listing records before accession deduplication. I report missing prices before the USD 3 screen.

Retained condition	Remaining	Removed
Eligible SEC filings in 2021-2025	1683	0
Downloaded and parsed	1683	0
Unique accession	1663	20
Required forms and filing dates	1663	0
At least 2,000 parsed tokens (both forms)	1663	0
Observed acceptance and tradable day 0	1663	0
Unambiguous exact-filing cover-page shares	1070	593
Observed unadjusted previous-session close	1053	17
Unadjusted previous-session close at least USD 3	1007	46
Complete four-day stock and SPY returns	1007	0
Complete 55-day pre-filing return window	1002	5
Complete 63-day post-filing return window	1002	0
Complete positive pre-filing volume control	1002	0
Finite size and turnover controls	1002	0

The 2,000-token and USD 3 thresholds are research choices. I require a common complete-case sample so the paired regressions use the same observations and the same IDF corpus.

2. Language measures

I use the Loughran-McDonald financial dictionary: 2,355 negative words and 297 uncertainty words, with 40 in both lists. Negative tone describes adverse outcomes; uncertainty describes qualifications and indeterminacy. A financial dictionary avoids the broad negative labels that general-purpose lists assign to ordinary accounting language.

Table 2. Summary statistics by report type

form	measure	count	mean	std	25%	50%	75%
10-K	Neg-P	248	0.02466	0.003747	0.02192	0.02498	0.02737
10-K	Neg-T	248	175.7	67.83	124.2	153.6	224.4
10-K	Unc-P	248	0.0208	0.002553	0.01907	0.02098	0.02265
10-K	Unc-T	248	28.75	9.77	21.38	26.46	34.15
10-Q	Neg-P	754	0.01971	0.009613	0.01178	0.0161	0.02873
10-Q	Neg-T	754	66.58	71.1	17.46	30.27	105
10-Q	Unc-P	754	0.01842	0.006441	0.01369	0.0166	0.02403
10-Q	Unc-T	754	11.91	8.464	5.669	9.334	14.89

P denotes the fraction of tokens in the category; T denotes the summed tf.idf weight. Neg is negative language and Unc is uncertainty. The notebook includes minima, maxima and the correlation matrix.

The negative/uncertainty proportion correlation is 0.736 for 10-K; 0.903 for 10-Q.

Table 3. Thirty most frequent words in each list

Rank	Negative word	% of Neg	Uncertainty word	% of Unc
1	LOSS	4.892	MAY	37.03
2	ADVERSELY	4.026	COULD	19.45
3	ADVERSE	3.285	RISK	5.675
4	CLAIMS	2.751	RISKS	5.025
5	AGAINST	2.198	BELIEVE	2.929
6	UNABLE	2.167	APPROXIMATELY	2.189
7	LOSSES	2.083	ASSUMPTIONS	1.496
8	FAILURE	2.022	UNCERTAINTIES	1.233
9	HARM	1.811	FLUCTUATIONS	1.212
10	LITIGATION	1.711	ANTICIPATED	1.084
11	CRITICAL	1.379	INTANGIBLE	1.073
12	FAIL	1.37	POSSIBLE	1.067
13	NEGATIVELY	1.363	DEPEND	1.044
14	DELAYS	1.306	PREDICT	1.014
15	DELAY	1.182	UNCERTAIN	0.9607
16	DIFFICULT	1.144	UNCERTAINTY	0.9225
17	PENALTIES	1.109	ANTICIPATE	0.9112
18	NEGATIVE	1.039	MIGHT	0.7794
19	DECLINE	0.8785	DIFFER	0.7785
20	IMPAIRMENT	0.87	VOLATILITY	0.7425
21	CHALLENGES	0.8532	DEPENDS	0.6922
22	DISRUPTIONS	0.8483	EXPOSURE	0.6507
23	LIMITATIONS	0.7876	DEPENDENT	0.6172
24	RESTATED	0.7837	PENDING	0.6016
25	FINES	0.7045	VARY	0.5094
26	DAMAGES	0.6882	CONTINGENT	0.4751
27	HARMED	0.6639	FLUCTUATE	0.4617
28	VOLATILITY	0.6339	REVISED	0.452
29	DAMAGE	0.6293	CONTINGENCIES	0.4275
30	INABILITY	0.6264	DEPENDING	0.3654

The top 30 words account for 45.81% of negative occurrences and 91.87% of uncertainty occurrences. MAY alone accounts for 37.03% of uncertainty words. I use all category occurrences as the denominator. Common hedges dominate the proportions, which motivates the tf.idf comparison.

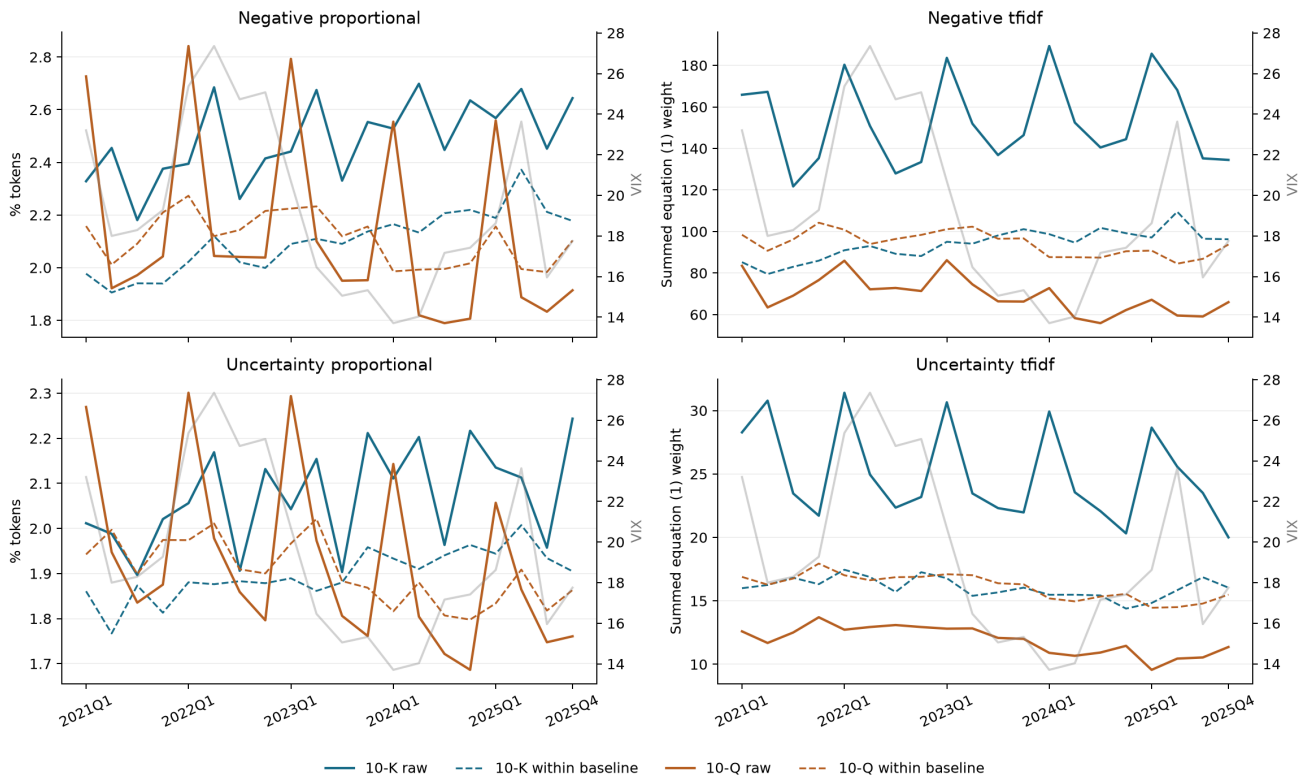
3. Measurement and quarterly composition

I retain the starter parser: it removes inline-XBRL elements, hidden content and tables with digits exceeding 15% of non-space characters. It produces uppercase tokens of at least two letters, allowing internal apostrophes and hyphens. I analyse the resulting whole-document text. The parser can remove narrative inside XBRL tags and leave boilerplate; the scores inherit these measurement limits.

Proportional tone is category word occurrences divided by total tokens. For each observed term, equation (1) of Loughran and McDonald gives $[(1 + \ln tf)/(1 + \ln a)] \ln(N/df)$, where a is total tokens divided by distinct observed tokens in that filing. I sum these weights within each category. N and df use the final regression corpus. This full-sample construction contains future corpus information, so I interpret the tests as retrospective associations.

Figure 1. Quarterly language and VIX

Figure 1. Filing language by calendar quarter; grey line is VIX



I average multiple filings within each firm-form-quarter, giving 1,000 cells from 1,002 filings, and then give each cell equal weight. Solid lines separate 10-Ks and 10-Qs. For the dashed lines I subtract each firm-form mean and add the pooled mean; these lines describe baseline composition in an unbalanced panel. I use fixed effects in Table 4 for the trend test. The grey right-axis series is the quarterly mean VIX.

Annual reports cluster in particular quarters and contain more risk discussion. Firms also differ in their baseline language. I therefore separate report types and include quarter-of-year controls in both aggregate and within-firm trends. I use VIX as context; the overlay alone cannot establish a predictive relation.

4. Trends within firms

Neg-P: the within-firm-form slope is -2.512×10^{-5} per quarter (95% CI $[-0.0001309, 8.063 \times 10^{-5}]$; clustered $p=0.625$).

Neg-T: the within-firm-form slope is -0.3041 per quarter (95% CI $[-1.007, 0.3991]$; clustered $p=0.377$).

Unc-P: the within-firm-form slope is -4.573×10^{-5} per quarter (95% CI $[-0.0001058, 1.439 \times 10^{-5}]$; clustered $p=0.128$).

Unc-T: the within-firm-form slope is -0.1419 per quarter (95% CI $[-0.2314, -0.05236]$; clustered $p=0.00362$).

I find the clearest pooled trend in weighted uncertainty. The form-specific estimates distinguish rising proportional negativity and uncertainty in 10-Ks from declining uncertainty in 10-Qs. Aggregate 10-Q negativity crosses the 5% threshold under OLS and Newey-West, whereas the within-firm estimate does not. I therefore avoid reading the aggregate decline as a change shared by firms.

Table 4. Aggregate and within-firm trend tests

form	measure	Estimator	coefficient	t	p	n
10-K	Neg-P	Aggregate OLS	0.00016	8.543	3.798×10^{-7}	20
10-K	Neg-P	Aggregate NW(4)	0.00016	8.817	2.549×10^{-7}	20
10-K	Neg-T	Aggregate OLS	0.6294	2.242	0.04047	20
10-K	Neg-T	Aggregate NW(4)	0.6294	3.599	0.002631	20
10-K	Unc-P	Aggregate OLS	8.094×10^{-5}	4.353	0.0005681	20
10-K	Unc-P	Aggregate NW(4)	8.094×10^{-5}	4.391	0.0005263	20
10-K	Unc-T	Aggregate OLS	-0.1188	-1.787	0.09411	20
10-K	Unc-T	Aggregate NW(4)	-0.1188	-2.707	0.01622	20
10-Q	Neg-P	Aggregate OLS	-0.0001208	-3.548	0.002921	20
10-Q	Neg-P	Aggregate NW(4)	-0.0001208	-2.938	0.01017	20
10-Q	Neg-T	Aggregate OLS	-0.8441	-4.216	0.0007478	20
10-Q	Neg-T	Aggregate NW(4)	-0.8441	-3.679	0.002231	20
10-Q	Unc-P	Aggregate OLS	-9.846×10^{-5}	-4.815	0.000227	20
10-Q	Unc-P	Aggregate NW(4)	-9.846×10^{-5}	-4.259	0.0006865	20
10-Q	Unc-T	Aggregate OLS	-0.1556	-5.81	3.437×10^{-5}	20
10-Q	Unc-T	Aggregate NW(4)	-0.1556	-4.68	0.0002962	20
Pooled	Neg-P	Within, 2-way	-2.512×10^{-5}	-0.4971	0.6248	1000
Pooled	Neg-T	Within, 2-way	-0.3041	-0.9051	0.3768	1000
Pooled	Unc-P	Within, 2-way	-4.573×10^{-5}	-1.592	0.1278	1000
Pooled	Unc-T	Within, 2-way	-0.1419	-3.317	0.003623	1000
10-K	Neg-P	Within, 2-way	0.0001558	8.152	1.265×10^{-7}	248
10-K	Neg-T	Within, 2-way	0.8523	3.392	0.003058	248
10-K	Unc-P	Within, 2-way	6.196×10^{-5}	6.081	7.566×10^{-6}	248
10-K	Unc-T	Within, 2-way	-0.1152	-1.793	0.08897	248
10-Q	Neg-P	Within, 2-way	-8.369×10^{-5}	-1.436	0.1674	752
10-Q	Neg-T	Within, 2-way	-0.6734	-1.783	0.09057	752
10-Q	Unc-P	Within, 2-way	-7.968×10^{-5}	-2.323	0.0314	752
10-Q	Unc-T	Within, 2-way	-0.1511	-3.245	0.004258	752

Aggregate regressions use 20 quarter means and quarter-of-year effects. Newey-West uses four lags with a finite-sample correction. Within regressions include firm-form and quarter-of-year effects; I cluster covariance by CIK and filing quarter and use t inference with the smaller cluster count minus one. In the 10-K sample, the Q3 indicator duplicates one firm effect. The least-squares fit identifies the time slope, but cannot separate those two nuisance effects. I report the full specification set without multiple-testing adjustments. Marginal p values warrant caution.

5. Uncertainty and subsequent volatility

Day 0 is the first NYSE session whose close follows SEC acceptance, using the actual close on early-close days. This moves 543 retained filings relative to their filing date. I compute annualised sample standard deviations of adjusted daily returns over [-60,-6] and [1,63]: 55 pre-filing and 63 post-filing sessions. The post window approximates the following trading quarter and can overlap later filings.

I regress log post-volatility on uncertainty, log size, log turnover, log token count, pre-filing excess return, form and filing-quarter effects. The paired specification adds log pre-volatility on the same rows. I standardise tone on the common sample. Size uses the same filing's cover-page shares and the previous-session price; I align historical price, volume and share units with observed split ratios.

Table 5. Uncertainty coefficients with and without pre-volatility

form	weighting	Pre-vol?	coefficient	se	p	n
Pooled	proportional	False	0.1043	0.04145	0.02105	1002
Pooled	proportional	True	0.04424	0.01853	0.02749	1002
Pooled	tfidf	False	0.08383	0.05729	0.1598	1002
Pooled	tfidf	True	0.0402	0.02409	0.1115	1002
10-K	proportional	False	0.1498	0.07412	0.05758	248
10-K	proportional	True	0.0695	0.0698	0.3319	248
10-K	tfidf	False	0.142	0.06579	0.04391	248
10-K	tfidf	True	0.07075	0.03666	0.06866	248
10-Q	proportional	False	0.1189	0.04735	0.02122	754
10-Q	proportional	True	0.05385	0.02193	0.02386	754
10-Q	tfidf	False	0.06066	0.09035	0.51	754
10-Q	tfidf	True	0.02453	0.03383	0.4773	754

For proportional uncertainty, adding pre-volatility changes the coefficient from 0.1043 ($p=0.021$) to 0.04424 ($p=0.0275$), a reduction of 57.6%. The controlled 95% interval is [0.005461, 0.08302].

For tfidf uncertainty, adding pre-volatility changes the coefficient from 0.08383 ($p=0.16$) to 0.0402 ($p=0.112$), a reduction of 52.0%. The controlled 95% interval is [-0.01022, 0.09062].

The attenuation is the central result: earlier volatility accounts for much of the raw uncertainty association. The proportional measure retains an incremental association at 5%; the weighted measure has a wider interval that includes zero. These observational regressions do not establish causation.

6. Precision of the four-day return test

Form	Weight	N	SE, return pp	80% MDE, pp
Pooled	proportional	1002	0.2848	0.8356
Pooled	tfidf	1002	0.7581	2.225
10-K	proportional	248	1.843	5.41
10-K	tfidf	248	1.491	4.376
10-Q	proportional	754	0.3311	0.9717
10-Q	tfidf	754	0.8034	2.358

The approximate two-sided 5%, 80%-power minimum detectable effect is (t critical + 0.842) times the clustered SE, per one-SD tone change. I express it in return percentage points. This sensitivity calculation uses estimated noise; it leaves smaller effects unresolved.

Table 6. Negative tone and four-day excess returns

form	weighting	coefficient	se	p	ci_low	ci_high
Pooled	proportional	-0.003198	0.002848	0.2754	-0.009158	0.002762
Pooled	tfidf	-0.004318	0.007581	0.5756	-0.02019	0.01155
10-K	proportional	-0.005743	0.01843	0.7588	-0.04433	0.03284
10-K	tfidf	0.002805	0.01491	0.8528	-0.0284	0.03401
10-Q	proportional	-0.006888	0.003311	0.05129	-0.01382	4.254e-05
10-Q	tfidf	-0.006871	0.008034	0.4031	-0.02369	0.009946

I subtract the compounded SPY return from the compounded stock return over [0,3]. Coefficients are return fractions per one-SD negative tone. Controls match Table 5 with log pre-volatility included. For intraday filings, close-to-close returns include some price movement before publication, which limits a trading interpretation.

The pooled proportional coefficient is -0.320 return percentage points (95% CI [-0.916, 0.276]; p=0.275).

The pooled tfidf coefficient is -0.432 return percentage points (95% CI [-2.019, 1.155]; p=0.576).

The 10-Q proportional estimate has p=0.0513. Its interval includes zero at 5%, and I do not count it as evidence of predictive returns. The broad pooled intervals remain compatible with small negative effects. I regard the return test as inconclusive; that judgment does not imply that the trend and volatility tests lack power.

7. Interpretation and limitations

Mean negative and uncertainty fractions are 2.47% and 2.08% for 10-Ks, versus 1.97% and 1.84% for 10-Qs. Different report scopes and repetition offer plausible explanations for these gaps. A difference between significance labels would require a formal interaction test before I could interpret it as a difference between slopes.

The complete-case filters retain 1,002 of 1,683 filing-listing records from 60 firms. The frozen holdings mix dates labelled 01/02/2026 and 09/04/2026. They omit companies ARK sold before those dates. I cannot measure that survivorship effect from this snapshot. The share-count requirement also removes firms with missing or ambiguous flat-API facts, including multi-class issuers.

SPY proxies for the broad market, and cover-page shares may predate the price used for size. Twenty quarters limit inference about common shocks. Two-way covariance produces undefined standard errors for 101 of 468 control or fixed-effect entries; I leave those entries undefined and do not interpret them. The reported tone and trend standard errors are finite. The parser, sample selection and retrospective IDF remain sources of uncertainty.

I place the most weight on the within-firm decline in weighted uncertainty and the attenuation of the volatility coefficient after controlling for pre-volatility. Historical holdings including exits and a chronological holdout with training-only IDF would help test whether these associations generalise.

Sources and reproduction

Loughran, T. and B. McDonald (2011), When Is a Liability Not a Liability?, Journal of Finance 66, 35-65, doi:10.1111/j.1540-6261.2010.01625.x. Inputs: SEC EDGAR, the instructor's frozen ARK holdings and LM dictionary, and Yahoo Finance through yfinance. The repository contains the executed notebook, scripts, identity configuration, results, environment instructions and AI_USE.md. Supply the personal repository URL with the Brightspace PDF.

Appendix to Table 1. Excluded holdings

One row per original security. A blank CIK means that I did not verify a SEC issuer match. For matched issuers I checked 2021-2025 form counts in SEC submissions. The notebook and src/issuer_map.py provide identity evidence links. Discovery, LY Corporation and JD Logistics remain explicit matching exclusions; I do not substitute a similarly named US company or a parent.

Original code	Holding issuer	CIK	Reason
2618	JD LOGISTICS INC		Foreign listing; no verified SEC issuer match
4689	LY CORP		Foreign listing; no verified SEC issuer match
6301	KOMATSU LTD	0000056594	No 10-K/10-Q; other forms or no filings in 2021-2025
ADYEN	ADYEN NV	0001788707	No 10-K/10-Q; other forms or no filings in 2021-2025
AIR	AIRBUS SE	0001697546	No 10-K/10-Q; other forms or no filings in 2021-2025
ALMR	ALAMAR BIOSCIENCES INC	0002104204	No 10-K/10-Q; other forms or no filings in 2021-2025
ARKY	ARK ACTIVE AUTOCALL INC ETF		Fund security
BABA	ALIBABA GROUP HOLDING-SP ADR	0001577552	No 10-K/10-Q; 20-F/6-K
BIDU	BAIDU INC - SPON ADR	0001329099	No 10-K/10-Q; 20-F/6-K
BLSH	BULLISH	0001872195	No 10-K/10-Q; 6-K
BYDDY	BYD CO LTD-UNSPONSORED ADR	0001445162	No 10-K/10-Q; other forms or no filings in 2021-2025
CBRS	CEREBRAS SYSTEMS INC - A	0002021728	No 10-K/10-Q; other forms or no filings in 2021-2025
CCJ	CAMECO CORP	0001009001	No 10-K/10-Q; 40-F/6-K
DSY	DISCOVERY LTD		Foreign listing; no verified SEC issuer match
DSY FP	DASSAULT SYSTEMES SE	0001016118	No 10-K/10-Q; other forms or no filings in 2021-2025
ESLT	ELBIT SYSTEMS LTD	0001027664	No 10-K/10-Q; 20-F/6-K
ETHQ/U	3IQ ETHER STAKING ETF		Fund security
ETOR	ETORO GROUP LTD-A	0001493318	No 10-K/10-Q; 6-K
FUTU	FUTU HOLDINGS LTD-ADR	0001754581	No 10-K/10-Q; 20-F/6-K

Appendix to Table 1. Excluded holdings (continued)

One row per original security. A blank CIK means that I did not verify a SEC issuer match. For matched issuers I checked 2021-2025 form counts in SEC submissions. The notebook and src/issuer_map.py provide identity evidence links. Discovery, LY Corporation and JD Logistics remain explicit matching exclusions; I do not substitute a similarly named US company or a parent.

Original code	Holding issuer	CIK	Reason
GENB	GENERATE BIOMEDICINES INC	0002100782	No 10-K/10-Q; other forms or no filings in 2021-2025
GENI	GENIUS SPORTS LTD	0001834489	No 10-K/10-Q; 20-F/6-K
GLBE	GLOBAL-E ONLINE LTD	0001835963	No 10-K/10-Q; 20-F/6-K
HO	THALES SA	0001884735	No 10-K/10-Q; other forms or no filings in 2021-2025
KLAR	KLARNA GROUP PLC	0002003292	No 10-K/10-Q; 6-K
KMTUY	KOMATSU LTD -SPONS ADR	0000056594	No 10-K/10-Q; other forms or no filings in 2021-2025
KSPI	JSC KASPI.KZ ADR	0001985487	No 10-K/10-Q; 20-F/6-K
NU UN	NU HOLDINGS LTD/CAYMAN ISL-A	0001691493	No 10-K/10-Q; 20-F/6-K
PONY	PONY AI INC	0001969302	No 10-K/10-Q; 20-F/6-K
PRNT	THE 3D PRINTING ETF		Fund security
SCTX	SCRIBE THERAPEUTICS INC	0001853921	No 10-K/10-Q; other forms or no filings in 2021-2025
SE	SEA LTD-ADR	0001703399	No 10-K/10-Q; 20-F/6-K
SLMT	BRERA HOLDINGS PLC-CL B	0001939965	No 10-K/10-Q; 20-F/6-K
SOLQ/U	3IQ SOLANA STAKING ETF		Fund security
SPCX	SPACE EXPLORATION TECHN-CL A	0001181412	No 10-K/10-Q; other forms or no filings in 2021-2025
SPOT	SPOTIFY TECHNOLOGY SA	0001639920	No 10-K/10-Q; 20-F/6-K
TSM	TAIWAN SEMICONDUCTOR-SP ADR	0001046179	No 10-K/10-Q; 20-F/6-K
WRD	WERIDE INC-ADR	0001867729	No 10-K/10-Q; 20-F/6-K
XE	X-ENERGY INC	0002088896	No 10-K/10-Q; other forms or no filings in 2021-2025